ZERO-SHOT VISION- LANGUAGE MODEL APPLICATION

AN INTERNSHIP PROJECT REPORT

Submitted by

SARASWATHY M

(Reg No: 110725102221)

in partial fulfilment for the award of the degree of

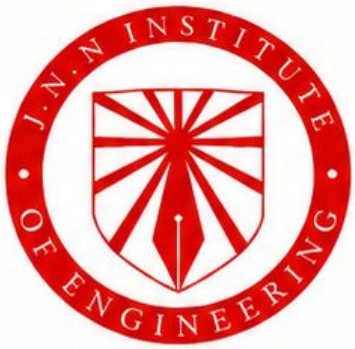
BACHELOR OF TECHNOLOGY

IN

Artificial Intelligence & Data Science

Internship carried out at

PENTA BRAINS SOLUTIONS PRIVATE LIMITED, CHENNAI



J.N.N Institute of Engineering

(An Autonomous Institute, Affiliated to Anna University, Chennai)

AUGUST – 2026

VISION OF THE INSTITUTE:

Lead the transformation of engineering and management learning experience to educate the next generation of innovators and entrepreneurs who want to make the world a better place.

MISSION OF THE INSTITUTE:

IM1: To develop the required resources and infrastructure and to establish a conducive ambience for the teaching-learning process.

IM2: To nurture professional and ethical values in the students and to instil in them a spirit of innovation and entrepreneurship.

IM3: To encourage a desire for higher learning and research in the students and to equip them to face global challenges.

IM4: To provide opportunities for students to learn job-relevant skills to make them industry ready.

IM5: To interact with industries and other organization to facilitate transfer of knowledge and know-how.

VISION OF THE DEPARTMENT:

To impart quality-education, inculcate professionalism and enhance the problem-solving skills of the students in the domain of Artificial Intelligence & Data Science by applying recent technological tools and incorporating collaborative principles with a focus to make them industry ready.

MISSION OF THE DEPARTMENT:

DM 1: To enhance the knowledge of the students with most recent advancements and refresh their insights in the field of Artificial Intelligence and Data Science.

DM 2: To equip the students with strong fundamental concepts, analytical capability, programming and problem-solving skills.

DM 3: To make the students industry ready and to enhance their employability through training, internships and real-time projects.

DM 4: To guide the students to perform research on Artificial Intelligence and Data Science, with the aim to provide solutions to the problems of the industry.

PROGRAM OUTCOMES

S.NO	GRADUATE ATTRIBUTE	PROGRAM OUTCOMES
PO1	Engineering knowledge	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis	Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/Development of solutions	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern Tool usage	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The Engineer and society	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

S.NO	GRADUATE ATTRIBUTE	PROGRAM OUTCOMES
PO7	Environment and sustainability	Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Individual and team work	Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication	Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM EDUCATIONAL OBJECTIVES (PEO)

S.NO	PROGRAM EDUCATIONAL OBJECTIVES (PEO)
PEO1	To provide graduates with the proficiency to utilize the fundamental knowledge of basic sciences, mathematics, Artificial Intelligence, data science and statistics to build systems that require management and analysis of large volume of data.
PEO2	To enrich graduates with necessary technical skills to pursue pioneering research in the field of AI and Data Science and create disruptive and sustainable solutions for the welfare of ecosystems.
PEO3	To enable graduates to think logically, pursue lifelong learning and collaborate with an ethical attitude in a multidisciplinary team.

PROGRAM SPECIFIC OBJECTIVES (PSOs)

S.NO	PROGRAM SPECIFIC OBJECTIVES (PSOs)
PSO1	Graduates should be able to evolve AI based efficient domain specific processes for effective decision making in several domains such as business and governance domains.
PSO2	Graduates should be able to arrive at actionable Foresight, Insight, hindsight from data for solving business and engineering problems.
PSO3	Graduates should be able to create, select and apply the theoretical knowledge of AI and Data Analytics along with practical industrial tools and techniques to manage and solve wicked societal problems.

BONAFIDE CERTIFICATE

Certified that this internship project report **" ZERO-SHOT VISION- LANGUAGE MODEL APPLICATION"** is the bonafide work of **"SARASWATHY M" (Reg No: 110725102221)**, who carried out the internship project work in the domain of **Application development** at **Penta Brains Solutions Private Limited, Chennai**, from 03.08.2026 to 14.08.2026 under my supervision.

Signature of the HOD with date

Dr. P. Nagarajan
Professor and Head,
Department of Artificial Intelligence & Data Science,
J.N.N Institute of Engineering, Chennai

Signature of the Supervisor with date

Miss. Brindha. S
Project Supervisor,
Department of Artificial Intelligence & Data Science,
J.N.N Institute of Engineering, Chennai

Certified that the above student has attended the viva voce examination held on .....

INTERNAL EXAMINER **EXTERNAL EXAMINER**

INTERNSHIP COMPLETION CERTIFICATE

Pentabrains

PENTABRAINS SOLUTIONS PRIVATE LIMITED

APPLICATION DEVELOPMENT INTERNSHIP & PROJECT COMPLETION CERTIFICATE

Certificate No.: PBS/CIPC/2026/022 **Date:08/21/2026**

CERTIFICATE OF COMPLETION

This is to certify that **SARASWATHY M** Register No. **110725102221,** a student of **J.N.N Institute of Engineering**, has successfully completed a 2 weeks Application Development Internship at Pentabrains Solutions Private Limited, Chennai, from 03/08/2026 to 14/08/2026.

Project Undertaken: "Zero-Shot Vision-Language Model Applications"

Key Areas Covered:

- Fundamentals of Vision-Language Models (VLMs)
- Object localization and grounding from natural language queries
- Zero-shot semantic segmentation using textual prompts
- Zero-shot visual question answering (VQA)
- One-shot learning with in-context instructions
- Prompt engineering for multimodal models
- Model evaluation and output visualization
- Best practices for deploying vision-language applications

Throughout the internship, **SARASWATHY M** completed the assigned project primarily through AI-assisted (vibe) coding rather than independent implementation. The internship objectives and the assigned project were completed to a basic standard.

We place this certificate on record to acknowledge the candidate's participation and completion of the internship period.

This certificate is issued on the basis of training and evaluation records maintained by Pentabrains Solutions Private Limited and may be verified by contacting the company directly.

For Pentabrains Solutions Private Limited,

PENTABRAINS SOLUTIONS PVT. LTD. CHENNAI

Janaki Raman K
Founder & CEO

Registered Office:
Pentabrains Solutions Private Limited
Plot No. 6A, Flat No. S1, Secretariat Colony,
Main Road, Lakshmipuruam, Kolathur, Chennai-99
suppport@pentabrains.com
www.pentabrains.com
+91 44 2565 0506

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First and foremost, I express my sincere gratitude to our Beloved **FOUNDER & CHAIRMAN Shri. S. JAYACHANDRAN B.Sc., B.L.,** and also heartfelt thanks to our **VICE CHAIRMAN Mr. NAVEEN JAYACHANDRAN M.S.,** for providing the necessary facilities for the completion of my project.

I thank our **PRINCIPAL Dr. G. MOHANBABU, Ph.D.,** for the encouragement provided with the congenial environment essential for me to complete the project.

I take this opportunity to express my gratitude to our **HEAD OF THE DEPARTMENT Dr. P. NAGARAJAN** and all other faculty members in the Department of ARTIFICIAL INTELLIGENCE AND DATA SCIENCE for giving me continuous encouragement for completing my project.

I would like to thank the management and mentors of Penta brains Solutions Private Limited for giving me the opportunity to do an Internship at Penta brains Solutions Private Limited I also would like all the people who worked with me with their patience and openness, they created an enjoyable working environment. It is indeed with a great sense of pleasure and immense sense of gratitude that I acknowledge the help of these individuals.

Finally, I would like to thank my parents and friends for their constant support and encouragement.

SARASWATHY M

ABSTRACT

This internship project focuses on the design and development of a web-based **Zero- Shot Vision-Language Model (VLM) Application** that integrates artificial intelligence, frontend development, backend programming, and database management. The main objective of the project was to develop an application capable of understanding and analysing images using natural-language prompts without requiring task-specific model training. The application provides a user-friendly frontend interface through which users can register, log in, manage their profiles, access a dashboard, and interact with the AI system through a chat-based interface. The project provided practical exposure to the integration of modern AI models with web application technologies and helped bridge the gap between theoretical learning and practical implementation.

The application was developed using **Python and Flask** for backend development, **PostgreSQL** for database management, **Moon dream 2B** for visual question answering and image description, and **CLIPSeg** for text-prompt-based image segmentation. Users can upload images and provide natural-language instructions to perform operations such as image description, product identification, product attribute analysis, and object segmentation. The system can identify product categories such as garments, footwear, bags, and accessories and analyse characteristics including colour, material, pattern, neckline, sleeve type, fit, logos, straps, and visible defects. The application also implements user authentication, profile management, password management, session handling, image uploading, and user activity streak tracking.

The frontend and backend components were integrated to provide an interactive platform where users can communicate with the AI models through a web interface. Flask handles the application routes and user requests, while the AI processing layer analyses uploaded images and generates the required results. CLIPSeg produces segmentation masks based on textual prompts, while Moondream performs image description, product categorization, and visual question answering

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CHAPTER 1
INTRODUCTION

1.1 OVERVIEW

Artificial Intelligence (Al) and Computer Vision have become important areas of modern software development, enabling computers to understand and process information from images and natural language. **Vision-Language Models (VLMs)** combine visual understanding with language processing, allowing users to interact with images through natural-language instructions. Unlike traditional computer vision systems designed for specific tasks, zero-shot VLM applications can perform different visual tasks based on textual prompts without requiring separate task-specific training

The project developed during the internship is a web-based **Zero-Shot Vision- Language Model Application** called **CHATLYTIX**. It provides an interactive platform for image analysis through a chat-based frontend. Based on the user's natural-language request, the system can perform image description, product identification, product attribute analysis, and object segmentation.

1.2 INTERNSHIP DETAILS

Table 1.1 – Details of the internship

Name of the Intern	SARASWATHY M
Register Number	110725102221
Institution	J.N.N Institute of Engineering (An Autonomous Institute, Affiliated to Anna University, Chennai)
Degree / Department	B.Tech – Artificial Intelligence & Data Science
Organization	Penta Brains Solutions Private Limited, Chennai
Internship Domain	Application development
Project Title	Student Management & Performance Analytics System
Internship Period	03.08.2026 to 14.08.2026
Certificate Number	PBS/CIPC/2026/022

1.3 ABOUT THE OGANIZATION

Penta Brains Solutions Private Limited is a technology-oriented organisation located at Lakshmipuram, Kolathur, Chennai-99, Tamil Nadu. As described in the internship report, the

organisation provides internship and training activities that give students practical exposure to modern software technologies, application development, artificial intelligence, and Vision-Language Models.

The internship focus included application development and **Zero-Shot Vision- Language Model applications**. The work involved learning Al-based image processing, developing software components, integrating Al models, and implementing a functional web application. The report describes the organisation's learning approach as including expert trainers, a collaborative environment, and opportunities to discuss ideas and present projects.

The company vision and mission statements were described in the source report as providing practical exposure to emerging technologies and encouraging innovative software and Al solutions, with hands-on learning in application development, artificial intelligence, and Vision-Language Models.

1.4 OBJECTIVES

The objectives of the project are:

- Develop a web-based application that accepts uploaded images and natural- language instructions.
- Integrate a Vision-Language Model for image description, visual question answering, product identification, and product attribute analysis.
- Implement text-prompt-based object segmentation using CLIPSeg .
- Develop frontend and backend components with user registration, login, dashboard, profile management, chat, and password-management functions.
- Connect the application to PostgreSQL for user-related and application data

1.5 PROJECT OVERVIEW

CHATLYTIX provides a single web interface for several image-analysis tasks. Moondream is used for image understanding, visual question answering, product analysis, and image description. CLIPSeg is used to process a text prompt and image to produce a segmentation mask. The Flask backend connects the frontend, database, uploaded images, and Al models. The application includes user registration, login, dashboard access, profile management, password management, chat interaction, and activity tracking.

CHAPTER 2
COMPONENTS AND TECHNOLOGIES

2.1 FRONTEND INTERFACE

The frontend is developed using HTML and CSS and provides a simple interactive interface. The application contains a home page, signup page, login page, dashboard, chat page, profile page, edit-profile page, and password-management page. The home page provides navigation to login and registration. The dashboard provides access to Al chat, profile information, total chats, daily streak, account status, and recent searches.

2.2 FLASK BACKEND

Flask is used as the backend framework. It manages application routes, user requests, sessions, templates, file uploads, authentication, and communication between the frontend and Al models. The routes described in the report include home, signup, login, dashboard, profile, edit profile, password management, logout, and chat functionality.

2.3 POSTGRESQL DATABASE

PostgreSQL stores and manages user and application data. The Python application uses the psycopg2 library to connect to the database. User details such as name, email address, password, login activity, and chat-related information can be stored and retrieved. The database also supports the dashboard by providing information such as total chats, recent searches, and activity streak

2.4 MOONDREAM VISION-LANGUAGE MODEL

Moondream is the primary Vision-Language Model used in the application. The report states that the Moondream 2B model is loaded using the Transformers library. It is used to

understand uploaded images and answer questions based on natural-language prompts, generate image descriptions, identify product categories, and analyse product attributes.

2.5 CLIPSEG IMAGE SEGMENTATION

CLIPSeg performs text-guided image segmentation. The user provides a natural- language prompt describing the object to be segmented. The application passes the image and text prompt to CLIPSeg, which generates segmentation probabilities. These probabilities are converted into a binary mask, resized to the original image dimensions, and applied to the image to produce the segmented result. The resulting image is saved in the application's upload directory

CHAPTER 3

METHODOLOGY AND SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The system comprises five main stages: user input, application processing, Al processing, database operations, and output display. The user enters a natural-language request and uploads an image through the CHATLYTIX interface. Flask receives the request and processes the uploaded image. Moondream handles image understanding, product identification, visual question answering, and product analysis, while CLIPSeg handles text-

based image segmentation. PostgreSQL manages user and application- related information. The processed response or segmented image is then displayed through the web interface.

User input and image upload → Flask web application → Moondream / CLIPSeg → Image processing and result → Web interface / PostgreSQL

3.2 WORKING METHODOLOGY

1. User Input: The user enters a request through the chat interface and uploads an image.
2. Request Analysis: The Flask backend receives the request and determines the required operation.
3. Image Understanding: Moondream processes the image for description, visual question answering, product identification, and attribute analysis.
4. Product Analysis: The system identifies categories such as garment,footwear, bag, accessory, or other, and performs category-specific analysis.
5. Image Segmentation: When segmentation is requested, CLIPSeg processes the image together with the text prompt and generates a segmentation mask.
6. Result: The application displays the generated response or segmented image to the user.

3.3 IMAGE PROCESSING

The uploaded image is opened and converted into RGB format before being processed by the Al models. For segmentation, the image and text prompt are passed to CLIPSeg. The model generates probabilities, which are converted into a mask. The mask is applied to the original image to create the segmented output, which is saved in the uploads folder.

3.4 AI MODEL INTEGRATION

Moondream is integrated for image understanding tasks including image descriptions, visual question answering, product identification, and product attribute analysis. CLIPSeg is integrated separately for text-prompt-based image segmentation. Both models are loaded using the Transformers library and processed using PyTorch.

3.5 SOFTWARE IMPLEMENTATION

The application was developed using Python and Flask for backend and web- application logic. PostgreSQL is used for database connectivity and management, while PyTorch and Transformers provide Al model functionality. PIL is used for image processing, and HTML and CSS are used for the user interface.

3.6 ALGORITHM FLOW

1. Start.
2. User logs in or registers.
3. User uploads an image and enters a request.
4. Flask receives the request.
5. Analyse the user request.
6. If image understanding is required, use Moondream.
7. If segmentation is requested, use CLIPSeg.
8. Process the image and generate the result.
9. Display the result to the user.
10. Repeat for the next request.

CHAPTER 4
IMPLEMENTATION

4.1 FRONTEND AND APPLICATION ROUTES

The CHATLYTIX application uses Flask routes to serve the home page, signup, login, dashboard, profile, edit-profile, password-management, logout, and chat functionality. The dashboard receives dynamic information from the backend, including the user's name, total chats, and activity streak. Navigation links allow users to move between Dashboard, Chat, Profile, and Logout sections

4.2 AI MODEL INTEGRATION

The project loads the Moondream model through the Transformers library using the model identifier recorded in the source report as "moondream/ moondream-2b-2025-04-14-bit". The report also identifies the CLIPSeg model as "CIDAS/ clipseg-rd64-refined". Uploaded images are opened with PIL and converted to RGB format. For segmentation, the text prompt and image are prepared using the CLIPSeg processor before model inference

4.3 CHAT AND PRODUCT ANALYSIS

The chat functionality allows a user to upload an image and enter a natural-language question or request. The application sends the image and question to Moondream and returns the generated response for display. For product analysis, the system first identifies a product

category. The source report lists garment, footwear, bag, and other as categories and describes further category-specific analysis.

Depending on the category and user request, the application can analyse visual characteristics including colour, material, pattern, gender, season, texture, neckline, sleeve type, collar, fit, closure, pockets, logos, straps, and visible defects.

4.4 POSTGRESQL INTEGRATION

The application uses psycopg2 to connect the Flask application to PostgreSQL. Database operations support user accounts, profile information, chat-related data, login activity, and dashboard information. The source report includes code segments demonstrating library imports, model loading and image processing, frontend routes, chat/result generation, and PostgreSQL query handling. Because the supplied PDF is image-based and some source-code text is not reliably extractable, the original source- code pages are retained in the appendix for visual verification.

CHAPTER 5
RESULTS AND DISCUSSION

5.1 TESTING OBJECTIVES AND METHODOLOGY

Testing was performed to verify the functioning of the frontend, Flask backend, Al models, image-processing module, chat functionality, and PostgreSQL database. The major functionalities were tested to ensure that the application could receive user input, process images, generate Al-based results, and display output through the web interface.

- Functional Testing: Verified the main application modules, includingfrontend navigation, Flask routes, chat, profile, and dashboard.
- Image Processing Testing: Tested image uploading and processing to ensure images were correctly opened and prepared for Al analysis.
- Al Model Testing: Tested Moondream for image description, visual question answering, product identification, and product analysis.
- Image Segmentation Testing: Tested CLIPSeg with text prompts to verify that requested objects could be identified and highlighted.
- Database Testing: Verified PostgreSQL connectivity and the retrieval and management of user-related application data.

5.2 CORE TEST CASES AND RESULT

Test Case	Expected Behaviour	Observed Result	Status
TC-01: User Interface and Routes	Home, Dashboard, Chat, Profile, and Logout pages should be accessible.	Navigation pages were accessed successfully.	Passed
TC-02: Image Processing and Al Analysis	Uploaded image should be processed and analysed using the Al model.	The Al model generated the required analysis or response.	Passed
TC-03: Product Identification and Segmentation	The application should identify supported product categories and process text-based segmentation requests.	Product categories were identified and CLIPSeg generated the requested segmented output.	Passed
TC-04: Chat and PostgreSQL Database	User requests should produce results and application information should be handled through PostgreSQL.	Chat and database operations worked as expected.	Passed

5.3 RESULT AND DISCUSSION

The testing results reported for CHATLYTIX indicate that the major functional modules operated successfully. The application was able to receive user input, process images, generate Al-based results, and display the output through the web interface. The integration of Moondream and CLIPSeg supports multiple image-analysis tasks through a single application, while Flask coordinates the user interface, database, uploaded images, and Al models.

The product-analysis function supports categories such as garments, footwear, bags, and accessories and can analyse visual characteristics according to the user request. Text- prompt-based segmentation demonstrates how a natural-language description can be used to identify and highlight an object in an image. PostgreSQL supports the user- related and application data used by account and dashboard features.

The supplied report describes the application's implemented features and reports successful testing of its major functional modules. It does not provide quantitative performance metrics, a formal user-study result, deployment details, or model-accuracy measurements. These details are therefore not added to this report.

CHAPTER 6
CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The CHATLYTIX project demonstrates the development of a web-based Vision-Language Model application by integrating artificial intelligence, image processing, web development, and database technologies. The application provides users with an interactive platform for uploading images and performing image description, visual question answering, product identification, product analysis, and text-prompt-based image segmentation.

The application integrates Flask for backend development, HTML and CSS for the frontend, PostgreSQL for database management, Moondream for image understanding, and CLIPSeg for image segmentation. It also includes user authentication, dashboard functionality, chat interaction, profile management, and activity tracking. The reported testing results showed that the major functional modules operated successfully.

6.2 LEARNING OUTCOMES

The internship provided practical experience in integrating Vision-Language Models with a complete web application and helped develop knowledge in artificial intelligence, computer vision, image processing, database connectivity, backend development, frontend development, and prompt-based image analysis. It also provided experience in connecting pretrained Al models with a functional application interface

6.3 FUTURE WORK

CHATLYTIX can be improved in the future through the following enhancements:

- Improve image description, product identification, and analysis accuracy.
- Enhance CLIPSeg for more precise image segmentation.
- Support additional image categories and improve chatbot functionality.
- Optimize processing speed and deploy the application on a cloud platform.